

NCTL + CTBI (Scientific Proof)

1. Nighttime Baseline

Reused: **36.34 UTCI** (90th percentile of 6-year nighttime UTCI — matches Khushi's Day 2 calculation exactly, cross-validated).

2. NCTL — Nighttime Cumulative Thermal Load

Formula: $NCTL(t) = \sum \max(0, UTCI_{night}(h) - 36.34) \times \Delta h$, computed per calendar night (20:00 of one day through 05:59 of the next, treated as one continuous "night").

Result: 552 nights processed across 2021–2026. NCTL range: 0 to 12.25 degree-hours.

Graph: `nctl_graph.png` — clearly shows 6 distinct seasonal peaks (one per year), each showing the expected buildup during the hottest weeks of each season.

Note on graphing: dates across different years have real gaps between them (no data exists for July–March). Initially plotted as one continuous line, this created a false connecting line across year boundaries — fixed by inserting break points at each gap so the line doesn't imply continuity where none exists. Purely a visualization fix — the underlying NCTL values were never affected, since NCTL has no day-to-day dependency (each night is computed independently).

3. CTBI — Cumulative Thermal Burden Index (Per-Year Reset)

Formula: $CTBI(t) = \alpha \cdot CTBI(t-1) + \beta \cdot DayStress_{norm}(t) + \gamma \cdot NCTL_{norm}(t)$, with $\beta + \gamma = (1 - \alpha)$, split 50/50 between day and night contribution (documented prototype choice).

Critical implementation decision: CTBI resets to 0 at the start of each year's April–June window rather than running as one continuous 6-year series. This was necessary because ~9 months of missing data exist between each season — without the reset, the formula's

day-to-day memory term would wrongly treat June 30 and the following April 1 as consecutive days.

Main run ($\alpha = 0.7$): CTBI range 0.085 to 0.774 (normalized 0–1 scale). Graph: `ctbi_2024_graph.png` (2024 season) — shows the expected qualitative pattern: low at season start, rising through April–May, sustained high burden through peak summer, declining as the season ends.

4. Sensitivity Test ($\alpha = 0.6 / 0.7 / 0.8$)

Ran CTBI with all three α values on the same 2024 data. **Correlation between the $\alpha=0.6$ and $\alpha=0.8$ series: 0.947.**

Graph: `ctbi_sensitivity_graph.png` — all three lines track the same shape, just shifted slightly. This is the answer to "why 0.7?": *"0.7 is a prototype memory parameter, not a validated physiological constant. We tested the reasonable range 0.6–0.8 and confirmed the qualitative pattern — and our conclusions — remain stable across it."*

5. Multi-Year Validation (Bonus)

Graph: `ctbi_all_years_graph.png` — all 6 years' CTBI curves aligned by "days since April 1." Every year shows the same qualitative shape (low start, rising through the season, elevated through summer), confirming this isn't a pattern that only shows up in one cherry-picked year.

Files produced

- `nctl_ctbi.py` — reusable script (reproducible from raw UTCI + DayStress data)
- `nctl_graph.png`, `ctbi_2024_graph.png`, `ctbi_sensitivity_graph.png`, `ctbi_all_years_graph.png`
- `nctl.csv`, `ctbi_alpha_0.7.csv`, `ctbi_sensitivity.csv` (underlying data)

Handoff

- `ctbi_alpha_0.7.csv` (or the full sensitivity file) → **Garima**, for joining thermal values to ward polygons
- All 4 graphs → **Shweta**, for final presentation-quality polish and PPT integration

- This entire deliverable → **Khushi**, for independent sanity-checking against the expected scenarios (rising heat → CTBI rises, heat gone → CTBI declines, etc.)